

EVERY GRAPH WITH NO K_7^- MINOR IS SIX-COLOURABLE

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ABSTRACT. Let K_7^- be the graph obtained from K_7 by deleting one edge. We prove that every graph with no K_7^- minor is six-colourable. The main step is an extremal theorem: every 4-bilicht graph on $n \geq 3$ vertices with at least $4n - 2$ edges contains a K_7^- minor. In particular, this holds for all five-connected graphs. The key step strengthens a rooted density theorem of Dvořák, Norin and Rahman, reducing the number of missing adjacencies in their construction from two to one.

1. INTRODUCTION

All graphs in this paper are finite and simple. Write K_7^- for K_7 with one edge deleted. We prove the following statement, posed as Conjecture 21 by Norin and Totschnig [6].

Theorem 1.1. *Every graph with no K_7^- minor is six-colourable.*

Norin and Totschnig [6] proved the corresponding result when the excluded minor is K_7 with two incident edges deleted. Dvořák, Norin and Rahman [2] proved it for K_7 with two independent edges deleted. Theorem 1.1 strengthens both conclusions. It does not settle Hadwiger's conjecture for K_7 .

Our proof uses the rooted density and separation results of Dvořák, Norin and Rahman. Their critical-graph density bound, also obtained independently in our earlier work [3, Corollary 3], reduces Theorem 1.1 to an extremal problem. We prove the stronger density theorem stated as Theorem 2.1 below, for the class of 4-bilicht graphs. It gives the following result, conjectured in [2, Conjecture 1.5].

Corollary 1.2. *Every five-connected graph on $n \geq 6$ vertices with at least $4n - 2$ edges contains a K_7^- minor.*

The rooted construction in [2, Theorem 2.9] gives five branch sets containing prescribed roots and two additional branch sets. Of the eleven possible adjacencies involving the latter two sets, it allows two to be missing. Even when the five rooted branch sets are pairwise adjacent, this need not give K_7^- . Our rooted theorem allows only one missing adjacency when the rooted four-density is at least two. The difficulty is that an edge contraction may destroy the separation conditions needed for induction. We show that each such obstruction has a five-vertex boundary. A rooted-star construction handles the small configurations. For the rest, we add an auxiliary clique and apply Mader's atom theorem; its bound on the size of an atom contradicts a degree count. The auxiliary clique serves only this argument and is not used to construct a minor.

To prove the unrooted theorem, we must also deal with sides of five-cuts whose density is at most one. In a minimum counterexample, every five-cut has exactly one such side. Local contractions and rooted constructions exclude vertices of degree five; an inclusion-minimal fragment then excludes all five-cuts. Once six-connectivity is established, edge contractions force every neighbourhood to have minimum degree at least four. Minimality gives $e(G) = 4v(G) - 2$, so some vertex has degree six or seven. We finish with an adaptation of Norin–Totschnig's degree-six argument and a short complement classification for degree seven.

Section 2 gives the definitions and external results. The rooted theorem and the unrooted argument follow in Sections 4–7. Appendix A contains the degree-seven lemma.

2. PRELIMINARIES

2.1. Minors and densities. Write $v(G) = |V(G)|$, $e(G) = |E(G)|$ and $\rho(G) = e(G) - 4v(G)$. An H -minor model in G consists of pairwise disjoint, nonempty connected vertex sets $B_h \subseteq V(G)$, one for each $h \in V(H)$, such that B_h and $B_{h'}$ are adjacent whenever $hh' \in E(H)$. The sets B_h are the *branch sets* of the model. A graph (G, X) with a prescribed root set X is k -rooted if $|X| = k$. A rooted model contains each root in its corresponding branch set, and no branch set contains two roots. Additional edges between branch sets are allowed. We allow an empty root set and interpret its rooted clique K_0 as the empty model.

For a nonempty set $Y \subseteq V(G)$, let $N_G(Y)$ be its external neighbourhood, and put

$$\rho_G(Y) = e(G[Y]) + e_G(Y, N_G(Y)) - 4|Y|.$$

We call Y a *fragment*, its neighbourhood its *boundary*, and $V(G) \setminus (Y \cup N_G(Y))$ its *opposite side*. A fragment is *proper* if its opposite side is nonempty. In a rooted graph, fragments are required to avoid the roots. Set

$$\rho_4(G, X) = e(G) - e(G[X]) - 4|V(G) \setminus X|.$$

Thus $\rho_4(G[Y \cup N_G(Y)], N_G(Y)) = \rho_G(Y)$.

A k -rooted graph (G, X) is *4-light* if $\rho_G(Y) \leq 0$ for every root-free fragment whose boundary has fewer than k vertices. Thus, for five roots, the condition applies to boundaries of order at most four. The graph is *internally k -connected* if every nonempty root-free set has at least k external neighbours.

Two disjoint fragments with no edge between them form a *bifragment*; it is *dense* if both fragments have positive density. An unrooted graph is *4-bilight* if it has no dense bifragment with both boundary orders at most four. Every five-connected graph is 4-bilight. Our main extremal theorem is as follows.

Theorem 2.1. *Every 4-bilight graph G with $v(G) \geq 3$ and $e(G) \geq 4v(G) - 2$ contains a K_7^- minor.*

A separation (A, B) has $A \cup B = V(G)$ and no edge between $A \setminus B$ and $B \setminus A$; its order is $|A \cap B|$. It is proper if both open sides are nonempty. For a root separation we also require $X \subseteq A$. Its right-hand side is $G[B]$ rooted at $A \cap B$. When a linkage carries these boundary roots back to X , its paths are disjoint and their interiors avoid B . Roots already on the boundary use trivial paths. Appending each path to the corresponding branch set preserves disjointness and adjacencies.

When working in a minor, we fix a connected preimage for each vertex; these preimages are pairwise disjoint. A model in the minor lifts to the original graph by taking the union of the preimages in each branch set. This preserves connectivity, adjacencies and prescribed roots. Our minimal-counterexample arguments minimise $(v(G), e(G))$ lexicographically.

2.2. Rooted density theorems. Let $\mathcal{S}_{5,j}$ be the class of graphs on five vertices with at most j edges, and let $\mathcal{S}_{5,4}^-$ omit $K_3 \dot{\cup} K_2$ from $\mathcal{S}_{5,4}$. The following form of Dvořák's theorem [1, Theorem 4] is reproduced in [2, Theorem 2.6].

Theorem 2.2. *If (G, X) is a 4-light five-rooted graph and $t = \rho_4(G, X)$, then it has a rooted model of every labelled graph in $\mathcal{T}_t$, where*

t	≤ 0	1	2	3	4	5	6	≥ 7
$\mathcal{T}_t$	$\mathcal{S}_{5,0}$	$\mathcal{S}_{5,1}$	$\mathcal{S}_{5,3}$	$\mathcal{S}_{5,4}^-$	$\mathcal{S}_{5,6}$	$\mathcal{S}_{5,8}$	$\mathcal{S}_{5,9}$	$\mathcal{S}_{5,10}$

The labelling prescribes the root for each target vertex.

Corollary 2.3 ([2, Lemma 5.6]). *Let (G, X) be 4-light with five roots, and put $m = 10 - e(G[X])$. If $\rho_4(G, X) \geq \lceil (m + 3)/2 \rceil$, then G has a rooted K_5 minor.*

Proof. At this threshold and above, the target class contains every five-vertex graph with m edges. Apply Theorem 2.2 to the missing edges of $G[X]$, then retain the existing root edges. $\square$

A *dart* has four roots, two root edges forming a three-vertex path, and one nonroot adjacent to all four roots. The fourth root is isolated in the root-induced graph. Dvořák's small-root theorem [1, Corollary 13], together with [2, Lemma 3.1], implies that a 4-light k -rooted graph of positive rooted density contains a rooted K_k if $k \leq 3$, and a rooted dart if $k = 4$.

2.3. Reducible fragments and extraction. A fragment Y with boundary S of order at most four is *reducible* if $\rho_G(Y) \leq 0$ and $G[Y \cup S]$, rooted at S , contains either a rooted clique on S , or a rooted dart when $|S| = 4$ and $|Y| \geq 2$. For an unrooted ambient graph we also require $|V(G) \setminus Y| \geq 3$.

A *reducent* replaces Y by such a clique or dart, retaining the graph outside Y . The replacement is a minor: the rooted model provides disjoint connected preimages for the boundary vertices and, for a dart, its additional vertex. In both cases the order strictly decreases.

Lemma 2.4 ([2, Lemma 3.3]). *Replacing an inclusion-maximal reducible fragment preserves 4-bilighness and does not decrease ρ in an unrooted 4-biligh graph. In a 4-light five-rooted graph, it preserves 4-lightness and does not decrease ρ_4 . All original roots are retained in their prescribed, distinct branch sets.*

For the extraction step, two weaker patterns from [2] are useful. Each has five independent roots and two nonroots. In the first, the nonroots are adjacent and each misses at most one root, with distinct missed roots if both omissions occur. In the second, the nonroots are nonadjacent and both are adjacent to every root; this is $K_{2,5}$ rooted at its five-vertex part.

Lemma 2.5 ([2, Corollary 4.2]). *Let (G, X) be 4-light, with $|X| \leq 4$ and $\rho_4(G, X) \leq 0$. Suppose a root five-separation (A, B) has a right-hand side containing one of the two weaker patterns. Then G has a reducible fragment Y containing $B \setminus A$.*

We recall the extraction argument, which we also use for five roots. Restrict either pattern to a proper subset Z of its roots by deleting the other root vertices. The remaining graph contains a rooted clique if $|Z| \leq 3$, or a dart if $|Z| = 4$ [2, Observation 4.1]. If five disjoint paths do not link the original roots to the boundary, Menger's theorem gives an isolating separator of order at most four, linked to a proper subset of the boundary. The clique or dart lifts along this linkage. The isolated side has at least two interior vertices, since it contains both nonroot branch sets. Lightness gives nonpositive density when the separator is smaller than the root set. When its order equals the root count, take the separation $(X, V(G))$ instead; its density is $\rho_4(G, X) \leq 0$ by assumption. This gives the reducible fragment. To use it in an unrooted graph, we must also check that at least three vertices remain outside it.

We also use the root-forgetting rule [2, Observation 5.2]: if $|X| \leq 4$, (G, X) is 4-light and $\rho_4(G, X) \leq 0$, then the same is true after replacing X by any subset X' . To see the density bound, put $M = X \setminus X'$. For a set Y tested in the new rooted graph, or its entire nonroot set, put $Y' = Y \setminus M$. The old hypotheses give $\rho_G(Y') \leq 0$, interpreting the empty set as having density zero. At most $4|M \cap Y|$ additional incident edges are counted when $M \cap Y$ is restored: each such vertex has possible neighbours, outside Y' , only in $M \cap Y$ or $N_G(Y)$, and $|M| + |N_G(Y)| \leq 4$. Hence

$$\rho_G(Y) \leq \rho_G(Y') + 4|M \cap Y| - 4|M \cap Y| \leq 0.$$

3. LOCAL ROOTED CONSTRUCTIONS

We begin with three rooted constructions and a local contraction lemma, all used again in the unrooted argument.

The rooted-star lemma is related to [2, Lemma 4.10], which gives three spokes at one of five roots when $|V(H)| + |X| \leq 8$. Under the degree assumptions below, independence of X allows all $k - 1$ spokes and the bound $|V(H)| + |X| \leq 2k - 1$.

Lemma 3.1 (Rooted star). *Let $Z \subseteq V(H)$ have order k , and let $X \subsetneq Z$ be independent. Suppose that*

$$|V(H)| + |X| \leq 2k - 1, \quad d_H(v) \geq k - 1 \quad (v \notin X), \quad d_H(x) \geq k - |X| \quad (x \in X).$$

Then, for some $z \in Z \setminus X$, the set $(V(H) \setminus Z) \cup \{z\}$ is connected and adjacent to every vertex of $Z \setminus \{z\}$.

Proof. Put $T = V(H) \setminus Z$. A component D of $H[T]$ is adjacent to all but at most $|D|$ vertices of Z , since every vertex of D has at least $k - |D|$ neighbours in Z . Thus at most $|T|$ roots have no neighbour in some component of $H[T]$. Since $|T| + |X| \leq k - 1$, there is a vertex $z \in Z \setminus X$ adjacent to every component. Thus $T \cup \{z\}$ is connected, including when $T = \emptyset$.

A vertex $r \in Z \setminus \{z\}$ with no neighbour in T must be adjacent to z . For $r \notin X$ this follows from $d_H(r) \geq k - 1$; for $r \in X$, independence of X and $d_H(r) \geq k - |X|$ force adjacency to every vertex of $Z \setminus X$. $\square$

Lemma 3.2 (Rooted triangle). *Let (R, S) be internally three-connected, where $|S| = 3$. Suppose that $R - S$ is nonempty and connected, each root has a neighbour outside S , and every nonroot has degree at least four. Then R contains an S -rooted triangle minor.*

Proof. Three distinct vertices a, b, c of a two-connected graph have a rooted triangle minor. To see this, take a cycle through a, b and, if necessary, a two-fan from c to that cycle. Either their union contains a cycle through all three vertices, or it contains a theta whose three paths contain a, b, c separately. In the latter case, assign one end of the theta to the a -path and the other to the b -path; the interior of the third path is the branch set at c . These three branch sets are connected and pairwise adjacent.

Consider the block-cut tree of R , representing a root which is a cutvertex by its cutvertex node, and any other root by its block node. If the median of the three root nodes is a block node, the roots have three distinct attachments to that block. The block is two-connected, so the preceding observation supplies a triangle minor rooted at the attachments. Appending the three disjoint exterior root paths gives the result.

If the median is a cutvertex c , each component of $R - c$ contains at most one root. The nonroots in any such component would have boundary contained in c and the possible root, contrary to internal three-connectivity. Hence c is the only nonroot, but then its degree is at most three. This is also impossible. $\square$

We use the following terminal construction from the proof of [2, Lemma 3.1]. Assume that four independent roots are the leaves of a tree T . Its two degree-three branching vertices v_1, v_2 may coincide at a degree-four vertex. Choose T to minimise the length of the path between these vertices. If, after deleting v_1, v_2 , every partition of the roots into two pairs has a path joining the pairs, then the graph has a rooted dart. This part of the cited proof requires only these tree and path conditions; no density hypothesis is needed.

Lemma 3.3 (Rooted dart). *An internally four-connected graph with four prescribed roots, at least two nonroots, and nonroot minimum degree at least five contains a rooted dart.*

Proof. Write (F, Z) for the rooted graph and proceed by induction on $|V(F)|$. Delete the root-root edges; this preserves all nonroot degrees and boundaries. Every partition of Z into two pairs has two disjoint paths between the pairs. Otherwise Menger's theorem gives a separation of order at most one, with one pair on each side. The nonroots on either open side then have boundary contained in that pair and the separator, of order at most three. Internal four-connectivity would put all nonroots in the separator, contradicting their number.

Each component of $F - Z$ is adjacent to all four roots, so it supplies a tree whose leaves are exactly Z . Choose such a tree with a shortest branching path, with ends v_1, v_2 . Unless $\{v_1, v_2\}$ separates two roots from the other two, the terminal construction above gives a dart. The case $v_1 = v_2$ is covered, since a one-vertex separation of this kind was already excluded.

In the remaining case let (B_1, B_2) be the separation, with $B_1 \cap B_2 = \{v_1, v_2\}$ and two roots in each open side. Put

$$Z_i = (Z \cap B_i) \cup \{v_1, v_2\}, \quad W_i = B_i \setminus Z_i.$$

If both W_i are empty, the degree bound makes v_1, v_2 adjacent to each other and to every root. Use v_1 as the dart's helper and absorb v_2 into one root branch set: the resulting root star contains the required three-vertex path.

Otherwise choose a nonempty W_i . It has at least two vertices, since a single vertex would have at most the four neighbours in Z_i . Every subset of W_i has the same neighbours in $F[B_i]$ as in F . Thus $(F[B_i], Z_i)$ is internally four-connected and retains the nonroot degree bound. It omits the two opposite roots, so induction provides a Z_i -rooted dart.

Take the two disjoint paths between the two original root pairs established above. They cross the separator at different vertices. Their segments after their last visits to $\{v_1, v_2\}$ lie outside B_i , apart from their initial vertices, and end at the two opposite roots. Append these segments to the branch sets at v_1, v_2 . They are disjoint from the other branch sets and the helper, and restore the four original roots while preserving every required adjacency. $\square$

Call a graph *quasi-five-connected* if it is four-connected and has no four-separation with at least two vertices on each open side. The following local version of the crossing argument in [4, Lemma 1] does not assume contraction-criticality.

Lemma 3.4. *Let Q be five-connected, and let A have boundary $S = N_Q(A)$ of order five. Suppose that $|A| \geq 3$ and $|V(Q) \setminus (A \cup S)| \geq 2$. If $x \in S$ has a unique neighbour a in A , then Q/xa is quasi-five-connected.*

Proof. Since the graph is five-connected, an edge contraction preserves four-connectivity. A forbidden four-separation of Q/xa would lift to a five-cut T of Q containing x, a , with open sides B, D of order at least two. Write $C = V(Q) \setminus (A \cup S)$ and

$$p = |A \cap T|, \quad r = |C \cap T|, \quad s = |S \cap T|, \quad u = |S \cap B|, \quad w = |S \cap D|.$$

Then $p + r + s = u + w + s = 5$ and $s \geq 1$.

If $A \cap B \neq \emptyset$, its boundary is contained in $(A \cap T) \cup (S \cap T) \cup (S \cap B)$ and omits x , whose unique neighbour in A lies in T . Five-connectivity gives $p + s + u \geq 6$, so $p > w$ and $u > r$. The opposite corner $C \cap D$ has boundary of order at most $r + s + w < 5$, and is therefore empty. Symmetrically, $A \cap D \neq \emptyset$ implies $p > u$, $w > r$, and $C \cap B = \emptyset$.

If both A -corners are nonempty, then $C \subseteq T$, so $r \geq 2$, whereas $u + w \geq 2r + 2 \geq 6 > 5 - s$. If neither is nonempty, then $p = |A| \geq 3$ and $r \leq 1$. Some C -corner is nonempty, say $C \cap B$; its boundary gives $u \geq p$. Hence $w \leq 1$ and $C \cap D$ is empty, contrary to $|D| \geq 2$.

Finally suppose only $A \cap B$ is nonempty. Then $D = S \cap D$, so $w \geq 2$. Since $p > w$, we have $r \leq 1$. But $C \cap B$ has boundary of order at most $r + s + u = 5 + r - w \leq 4$, so it too is empty. This leaves $|C| = r \leq 1$, the final contradiction. $\square$

4. THE ROOTED THEOREM

For five prescribed roots X , say that a model has the *helper property* if it consists of five branch sets at the prescribed roots and two further nonempty connected branch sets, all pairwise disjoint, with at least ten of the eleven possible helper-root and helper-helper adjacencies. The two helpers contain no root; no root-root adjacency is required.

Theorem 4.1. *Every 4-light five-rooted graph (G, X) with $\rho_4(G, X) \geq 2$ has the helper property.*

We first establish structural properties of a minimum counterexample, then use a weighted neighbourhood to find an edge with few common neighbours. An auxiliary clique allows us to apply Mader's atom theorem without requiring intersections of positive-density fragments to have positive density.

4.1. Minimum-counterexample reductions. Suppose the theorem is false, and choose a counterexample minimising $(|V(G)|, |E(G)|)$ lexicographically. Throughout this section, the preimage of each vertex under a contraction contains at most one original root. A rooted model in a minor lifts by replacing each vertex with its fixed connected preimage; in particular, all five prescribed roots remain distinct.

We use the following consequence of the isolator argument in [2, Observation 4.1 and proof of Corollary 4.2]. Suppose a root-free region has a helper-property model on a five-vertex boundary, or two helpers adjacent to every branch set at a larger boundary. Either five disjoint exterior paths attach the original roots to five boundary branch sets, or a separator of order

$k \leq 4$ isolates the region and has a linkage to k of its boundary branch sets. A helper model restricted to those k branch sets contains a rooted K_k for $k \leq 3$, and a rooted dart for $k = 4$. Indeed, retain a helper adjacent to all four root branch sets and absorb the other into a root branch set to obtain two root–root adjacencies. For three roots, absorb the helpers into two root branch sets; smaller cases follow by restriction. The same conclusion holds when both helpers are adjacent to every branch set at a larger boundary. If the isolated interior has at least two vertices and nonpositive density, it is a reducible fragment. Apart from its ends, the linkage lies outside the original region and hence avoids the helpers. This is the only form of the isolator argument needed below.

Lemma 4.2. *The minimum counterexample has the following properties:*

- (1) X is independent, and there is no reducible root-free fragment with boundary of order at most four.
- (2) Every proper root-free set Y with $|N_G(Y)| = 5$ satisfies $\rho_G(Y) \leq 1$.
- (3) $G - X$ is connected, every nonroot has at most three root neighbours, and every root has degree at least two.
- (4) $\rho_4(G, X) = 2$.

Proof. Root–root edges contribute neither to the density nor to a required helper adjacency, and cannot occur inside a branch set containing just one root. Thus minimality makes X independent. An inclusion-maximal reducible fragment would give, by [2, Lemma 3.3], a smaller 4-light rooted minor of nondecreasing density. Its helper model would lift, proving (1).

For (2), put $S = N_G(Y)$. The induced rooted graph $(G[Y \cup S], S)$ is 4-light, since its root-free sets retain their incident edges and neighbours. If its density is at least two, it is a smaller graph with a helper model by minimality. Five exterior root paths would extend that model to X . Otherwise the isolator argument gives a reducible fragment: its interior contains the two helpers and has nonpositive density by 4-lightness. Both possibilities contradict (1).

The densities of the components of $G - X$ sum to $\rho_4(G, X)$. By 4-lightness, each component of positive density is adjacent to every root. Two such components give two helpers adjacent to all five roots, with only the adjacency between the helpers possibly missing. Hence there is exactly one component of positive density, and its density is at least two. Any other component would make this a proper five-boundary set, contrary to (2). Thus $G - X$ is connected and adjacent to every root.

Suppose a nonroot z has $d \geq 4$ root neighbours. For the components Q_i of $G - (X \cup \{z\})$,

$$\sum_i \rho_G(Q_i) = \rho_4(G, X) + 4 - d.$$

Each Q_i is adjacent to z , and each component of positive density is adjacent to at least four roots. If $d = 5$, there is a component of positive density; it and $\{z\}$ are adjacent helpers with at least ten required adjacencies. If $d = 4$, let x be the missing root. A component adjacent to every root already gives the required model. Otherwise each component of positive density has exactly four root neighbours, and has density at most one by (2). These densities sum to at least two, so at least two components have positive density. Choose a component R adjacent to x , and a component $Q \neq R$ of positive density. The helpers $\{z\} \cup R$ and Q are adjacent and have respectively five and at least four root adjacencies. This again contradicts the choice of G .

If a root x had just one neighbour v , the side $G - x$, rooted at $(X \setminus \{x\}) \cup \{v\}$, would have density $\rho_4(G, X) + 4 - |N_G(v) \cap X| \geq 3$. By 4-lightness, all five proposed boundary vertices must be actual neighbours of its interior. It is therefore a proper five-boundary side, contradicting (2). This proves (3).

Finally suppose $\rho_4(G, X) \geq 3$ and delete an edge with a nonroot end. The result has density at least two but, by minimality, is not 4-light. A positive small-boundary set Y becomes permissible only because the deleted edge joins Y to a vertex outside its closed side. Restoring that edge gives a five-boundary set of density at least two. By (2) its closed side is all of G , and its boundary is X . The restored root endpoint then has only one neighbour, contrary to (3). Therefore the density is exactly two. $\square$

Lemma 4.3. *Every nonroot has degree at least five, and (G, X) is internally five-connected.*

Proof. Deleting a nonroot v of degree at most four leaves density $2 + 4 - d_G(v) \geq 2$. If $G - v$ has a positive root-free set Y with at most four neighbours, restoring v must add a neighbour to Y ; otherwise Y already violates 4-lightness in G . Its restored density is at least two and its boundary has order five. This is a proper side, since its boundary contains the nonroot v and could not also contain all five roots. Lemma 4.2(2) excludes it. Thus deletion stays in the induction class, contradicting minimality.

If internal five-connectivity fails, choose a connected root-free set Y of minimum boundary order $k \leq 4$, and put $S = N_G(Y)$. The rooted graph $(G[Y \cup S], S)$ is internally k -connected, retains all nonroot degrees, and has nonpositive density. For $k = 0, 1, 2$, it contains a rooted K_k (using a path through Y when $k = 2$). For $k = 3$, apply Lemma 3.2. For $k = 4$, the degree bound implies $|Y| \geq 2$, and Lemma 3.3 applies. In every case Y is reducible, contrary to Lemma 4.2(1). $\square$

Complete X to a clique, obtaining $Q_0 = G + K_X$. It is five-connected: after deleting at most four vertices the surviving roots lie in one clique, and another component would contradict internal five-connectivity. For an original edge e having a nonroot end, let $t(e)$ be its number of common neighbours in Q_0 . The rooted density after contraction is

$$(1) \quad \rho_4(G/e, X_e) = 2 + 3 - t(e),$$

where X_e denotes the unchanged or naturally identified roots. The completion counts exactly the edges that become root-root edges when one endpoint is a root.

Lemma 4.4. *If $e = uv$ has a nonroot end and $t(e) \leq 3$, there is a proper root-free set Y such that*

$$|N_G(Y)| = 5, \quad \rho_G(Y) = 1, \quad \{u, v\} \subseteq N_G(Y),$$

and no vertex of Y is adjacent to both u and v .

Proof. By (1) and minimality, G/e is not 4-light. Let Y be a positive root-free set witnessing this. The identified vertex cannot lie outside its closed side, because then Y is a root-free set of boundary at most four in G . Nor can it lie inside Y : its preimage would have the same small boundary (and would be root-free; a contracted root cannot be interior). It therefore lies on the boundary. Splitting it gives exactly five neighbours in G . The side is proper, since its boundary contains a nonroot and hence cannot contain all five original roots. If c vertices of Y see both endpoints, then $\rho_G(Y) = \rho_{G/e}(Y) + c$. Lemma 4.2(2) forces both densities to equal one and $c = 0$. $\square$

Lemma 4.5. *There are no adjacent degree-five nonroots with exactly three common neighbours.*

Proof. Suppose p, q are such vertices. Write their common neighbours as the three-set A , and their respective exclusive neighbours as u, v . Lemma 4.4 applied to pq gives a density-one set T with boundary $\{p, q\} \cup D$, where $|D| = 3$, avoiding A . Since both p, q have neighbours in T , their neighbourhoods force $u, v \in T$; in particular these are nonroots.

Delete p, q from this side and reroot at $\{u, v\} \cup D$:

$$R = G[T \cup D], \quad Z = \{u, v\} \cup D.$$

The graph (R, Z) is 4-light. Indeed, every subset of its nonroots $T \setminus \{u, v\}$ has exactly the same neighbours and incident edges as in G , since only u, v in T see p, q . Put $a_0 = e_G(\{u, v\}, D)$ and $\delta = 1$ if $uv \in E(G)$, and $\delta = 0$ otherwise. Then

$$\rho_4(R, Z) = (4|T| + 1 - 2 - \delta - a_0) - 4(|T| - 2) = 7 - \delta - a_0.$$

This is exactly the number $m \in \{0, \dots, 7\}$ of missing edges from uv together with the six edges between $\{u, v\}$ and D . The rooted universality theorem [2, Theorem 2.6] supplies all these missing edges simultaneously: its table contains every m -edge graph in this range (the exceptional graph at density three has four edges). Retain the root edges already present. We obtain five rooted branch sets in R , with those at u, v adjacent to one another and to all three branch sets at D .

Append p to the branch set at u and q to the branch set at v . These give adjacent root-free helpers, each adjacent to every branch set at the boundary $A \cup D$. For $A \cap D$ use the existing branch sets at D ; for $A \setminus D$ use singletons. These branch sets are disjoint because $T \cap A = \emptyset$.

The root-free set $U = T \cup \{p, q\}$ has boundary exactly $A \cup D$. If five exterior paths link X to this boundary, retain the five branch sets at their distinct ends and append the paths. This gives the helper property rooted at X . Otherwise the isolator argument gives a separator of order at most four, a rooted clique or dart on it, and a root-free interior containing U . That interior has at least two vertices and nonpositive density, so is reducible. Both alternatives are impossible. $\square$

4.2. Eliminating degree five.

Lemma 4.6. *If u, v are nonroots and Q_0/uv is quasi-five-connected, then $(G/uv, X)$ is 4-light.*

Proof. Root completion does not change any root-free boundary or incident density. Every root has degree at least five in Q_0/uv : its at least two nonroot neighbours merge to at least one, and it retains the four other roots as neighbours. A root-free set with boundary at most four has nonempty opposite side, because there are five roots. Its boundary therefore has order four, and quasi-five-connectivity makes one open side a singleton. A singleton root-free side has degree four and density zero. A singleton opposite side would be a root of degree at most four, which is impossible. $\square$

Lemma 4.7. *Every nonroot of G has degree at least six.*

Proof. Suppose a nonroot v has degree five. First assume $Q_0[N_G(v)]$ is not complete, and choose a nonedge ab in it. Its ends are not both roots. The graph

$$H = G - v + ab$$

has rooted density two and is a proper rooted minor: contract va and delete the unwanted new edges. If H is 4-light, its helper model lifts, contrary to minimality.

Otherwise take a positive root-free set Y of H with at most four neighbours. Put $k = e_G(v, Y)$ and let c indicate whether the added edge ab is incident with Y . Then

$$N_G(Y) \subseteq N_H(Y) \cup \{v\}, \quad \rho_G(Y) = \rho_H(Y) + k - c.$$

Internal five-connectivity gives $|N_G(Y)| = 5$ and $k \geq 1$. This side is proper, since its boundary contains the nonroot v . Consequently

$$1 \geq \rho_G(Y) = \rho_H(Y) + k - c \geq 1 + k - c,$$

forcing $\rho_G(Y) = \rho_H(Y) = k = c = 1$. Exactly one endpoint of ab lies in Y ; call it a . Then a is a nonroot and the unique neighbour of v in Y . The other endpoint b is among the four vertices of $N_G(Y) \setminus \{v\}$, since the added edge cannot introduce a fifth neighbour in H .

The set Y is not a singleton, which would make ab an edge of G . If $|Y| = 2$, its density one and the nonroot degree bound force adjacent degree-five vertices with three common neighbours, contrary to Lemma 4.5. Hence $|Y| \geq 3$. Its opposite side in Q_0 has at least two vertices: a singleton would be an original root z , with boundary $(X \setminus \{z\}) \cup \{v\}$, and independence of X would give $d_G(z) \leq 1$.

Lemma 3.4 now applies to the unique edge va into Y . Thus Q_0/va is quasi-five-connected, and Lemma 4.6 makes G/va 4-light. Since v has degree five and b misses a , the edge va has at most three common neighbours. Its contraction retains rooted density at least two and decreases order, a contradiction.

It remains that $Q_0[N_G(v)]$ is a clique. Internal five-connectivity and Menger's theorem give five disjoint paths from X to $N_G(v)$, with interiors outside $N_G[v]$. Roots already in $N_G(v)$ use trivial paths. Choose $z \in N_G(v) \setminus X$, which exists because v has at most three root neighbours. Use $\{v\}, \{z\}$ as helpers, and the five linkage paths as root branch sets, removing z from its own path. The trimmed branch set is nonempty because z is not an original root. The helper z is adjacent to all five root branch sets, v is adjacent to the other four, and vz joins the helpers. Every clique edge used has nonroot end z , so is an edge of G . This gives the ten required adjacencies. $\square$

4.3. Weighted neighbourhoods and an atom. Let E_0 be the original edges with a nonroot end and at most three common neighbours in Q_0 . Each edge of E_0 has a five-boundary set as in Lemma 4.4.

Lemma 4.8. *If v is a nonroot and $d_G(v) + |N_G(v) \cap X| \leq 9$, then some edge of E_0 is incident with v .*

Proof. Suppose instead that $t(vu) \geq 4$ for every neighbour u of v . Put $H = G[N_G(v)]$, $X_0 = N_G(v) \cap X$, and $r = |X_0|$. Vertices of $H - X_0$ have degree at least four in H ; vertices of X_0 have degree at least $5 - r$, since root completion adds exactly $r - 1$ common neighbours to such an edge. Also X_0 is independent and $|V(H)| + r \leq 9$.

There are five disjoint paths from X to $N_G(v)$ in $G - v$. Otherwise a separator of order at most four would isolate, in G , a root-free component containing v . Truncate the paths at their first entries to $N_G(v)$. Their five ends form a set Z containing X_0 , and their interiors avoid the neighbourhood. Lemma 3.1 gives a central branch set

$$C = (N_G(v) \setminus Z) \cup \{z\}, \quad z \in Z \setminus X_0,$$

adjacent to the four other ends. Use C and $\{v\}$ as helpers, keeping the four other linkage paths and trimming z from its path. The trimmed root branch set is nonempty, and all branch sets are disjoint. The helper C has five root adjacencies, v has four, and they are adjacent. Thus the model has the helper property, a contradiction. $\square$

In particular, E_0 is nonempty. If $m = |V(G) \setminus X|$, the nonroot degree bound and at most three root neighbours give $m \geq 4$, while

$$(2) \quad \sum_{v \notin X} (d_G(v) + |N_G(v) \cap X|) = 8m + 4.$$

Some nonroot therefore has weight at most nine.

We use Mader's atom theorem in the form stated in [5, Theorem 5]. In a graph Q of connectivity k , let $\mathcal{S}$ be any family of vertex sets. An $\mathcal{S}$ -fragment is a nonempty proper union of components of $Q - T$, where T is a k -cut containing a member of $\mathcal{S}$. An $\mathcal{S}$ -atom is such a fragment of minimum order. For an atom A , put $T_A = N_Q(A)$ and $\bar{A} = V(Q) \setminus (A \cup T_A)$. If some $S \in \mathcal{S}$ and k -cut T satisfy

$$(3) \quad S \subseteq T \setminus \bar{A}, \quad A \cap T \neq \emptyset,$$

then

$$(4) \quad A \subseteq T, \quad |A| \leq \frac{1}{2}|T \setminus T_A|.$$

There is no assumption that every edge is noncontractible.

Completion of the proof of Theorem 4.1. Add a set W of $|V(G)| + 1$ vertices and make $X \cup W$ a clique, with no edges from W to $V(G) \setminus X$; call the resulting graph Q . Internal five-connectivity shows that Q is five-connected, and X separates $G - X$ from W , so its connectivity is exactly five. Adding this clique changes neither boundaries of original root-free sets nor common-neighbour counts of original edges with a nonroot end.

Let $\mathcal{S} = \{V(e) : e \in E_0\}$. By Lemma 4.4, $\mathcal{S}$ -fragments exist in Q , with an example of order at most $m = |V(G) \setminus X|$. Choose an $\mathcal{S}$ -atom A , so $|A| \leq m$. It avoids $X \cup W$: otherwise its five-boundary would force it to contain the whole surviving clique, of order at least $|W| > m$. Thus $A \subseteq V(G) \setminus X$, and its boundary is the same in G and Q .

Its closed side is proper in G . If not, its five-boundary would be exactly X , and could not contain the ends of an edge in E_0 . Lemma 4.2(2) therefore gives $\rho_G(A) \leq 1$. The nonroot degree bound excludes $|A| = 1, 2$, since for these orders

$$\rho_G(A) \geq 2|A| - \binom{|A|}{2} \geq 2.$$

Writing $S_A = N_G(A)$, double-counting incident edges gives

$$\sum_{v \in A} (d_G(v) + |N_G(v) \cap S_A|) = 8|A| + 2\rho_G(A) \leq 8|A| + 2.$$

As $|A| \geq 3$, some vertex $v \in A$ has the displayed weight at most eight. Its root weight is no larger, so Lemma 4.8 supplies an incident edge $f \in E_0$. Both ends of f lie in $A \cup T_A$.

Apply Lemma 4.4 to f , and let T be the boundary of its resulting fragment. In Q , this is a five-cut containing both ends of f and meeting A at v . Consequently $V(f) \subseteq T \setminus \bar{A}$, so (3) holds. Mader's theorem yields $|A| \leq 5/2$, contradicting $|A| \geq 3$.

We used the auxiliary clique only to select an atom; no model was constructed in the enlarged graph. The contradiction proves the theorem for the original rooted graph. $\square$

5. ORIENTING THE SEPARATIONS

Suppose that Theorem 2.1 fails, and choose a counterexample G minimising $(v(G), e(G))$ lexicographically. Thus G is 4-bilight, has at least three vertices and $\rho(G) \geq -2$, and has no K_7^- minor. Simplicity gives $v(G) \geq 9$. Throughout the next two sections, G denotes this minimum counterexample.

For a fragment A , write R_A for $G[A \cup N_G(A)]$ rooted at $N_G(A)$. We call a five-rooted side *heavy* if its rooted density is at least two, and *low* if its density is at most one. For a separation (A, B) , write $R_{A,B} = G[B]$, rooted at $A \cap B$.

Lemma 5.1. *The graph G has no reducible fragment.*

Proof. Replace an inclusion-maximal reducible fragment by a clique or dart rooted at its boundary. By Lemma 2.4, the resulting minor is 4-bilight and has density at least $\rho(G)$. The definition of reducibility leaves at least three vertices outside the fragment. The replacement strictly decreases the order: a clique removes every interior vertex, and a dart retains one where there were at least two. This contradicts minimality. $\square$

The next lemma adapts the separation argument of [2, Section 5]. We give the details to account for the density threshold of two and the final K_7^- model.

Lemma 5.2. *The following assertions hold.*

- (1) *If one side of a five-separation is heavy, the opposite side is 4-light.*
- (2) *For every five-separation with boundary X , exactly one side is heavy. If $m = 10 - e(G[X])$, the heavy side has density at least $m + 7$.*
- (3) *There is no bifragment (S, T) with both boundaries of order at most five, both densities positive, and every member with boundary of order five heavy.*

Proof. We first establish an observation about small separations. Suppose (C, D) has order at most four, $R_{C,D}$ is 4-light and has nonpositive density, and

$$|(C \setminus D) \cup N_G(C \setminus D)| \geq 3.$$

There cannot be a five-separation (A, B) with $B \subseteq D$ and $R_{A,B}$ both 4-light and heavy. To see this, remove from $C \cap D$ the vertices with no neighbour in $C \setminus D$, obtaining C' . The rooted graph $H = R_{C',D}$ remains 4-light and has nonpositive density, by the root-forgetting observation [2, Observation 5.2]: if a root set of order at most four is reduced, each forgotten root contributes at most four incident edges outside a given old fragment, offset by its density cost of four. The roots of H lie in A , so $(A \cap D, B)$ is a root separation of H with right side $R_{A,B}$. The helper model supplied there by Theorem 4.1 contains one of the rooted graphs in Lemma 2.5. That lemma gives a reducible fragment of H , which is also reducible in G : at least $|C'| \geq 3$ vertices remain outside it. This contradicts Lemma 5.1.

For (1), suppose that the side opposite a heavy side $R_{A,B}$ is not 4-light. It contains a positive fragment $L \subseteq A \setminus B$ with at most four neighbours. Bilightness implies that $R_{A,B}$ is 4-light, since a positive small-boundary fragment inside $B \setminus A$ would form a dense bifragment with L . Set

$$(C, D) = (L \cup N_G(L), V(G) \setminus L).$$

The side $R_{C,D}$ is 4-light and has nonpositive density: otherwise its whole interior, or a positive small-boundary fragment inside it, would again form a dense bifragment with L . Also $|L \cup N_G(L)| \geq 6$, since L has positive density. Now $B \subseteq D$, contradicting the preceding observation.

Let $a \geq b$ be the two densities at a five-separation with boundary X . Then

$$(5) \quad a + b = \rho(G) + 20 - e(G[X]) \geq m + 8.$$

In particular, $a \geq \lceil (m + 8)/2 \rceil$. If $b \geq 2$, (1) makes both sides 4-light. The rooted clique bound (Corollary 2.3) applies to the side of density a ; Theorem 4.1 applies to the other. Join corresponding root branch sets at their common root. The five resulting branch sets are pairwise adjacent; the two helper branch sets remain in the second open side and retain at least ten of their eleven possible adjacencies. This gives a K_7^- model. Thus $b \leq 1$, and (5) gives $a \geq m + 7$, proving (2).

For (3), first consider two heavy five-fragments S, T forming a bifragment. By (1), both R_S and R_T are 4-light. We claim that five disjoint paths join their closed sides. Otherwise Menger's theorem gives a separation (C, D) of order at most four containing $S \cup N_G(S)$ in C and $T \cup N_G(T)$ in D . Here $|C \setminus D| \geq 3$, since $|S| \geq 2$.

We verify that $R_{C,D}$ is 4-light and has nonpositive density. Discard from its root set the vertices with no neighbour in $D \setminus C$, leaving a separation (C, D') . No vertex of S can lie in $C \cap D'$, since all neighbours of S lie in C . Consequently $R_{C,D'}$ is a root side of order at most four in the side opposite R_S . By (1) it has nonpositive density, equal to that of $R_{C,D}$. Applying the same argument to every root side of $R_{C,D}$ proves its 4-lightness. The initial observation, applied to R_T , gives a contradiction. The five paths therefore exist.

Trim these paths so that they run between $N_G(S)$ and $N_G(T)$, with interiors outside both closed sides. The ends are distinct on each five-element boundary; a shared boundary vertex uses a trivial path. Apply the helper theorem in R_S and the rooted clique bound in R_T , using (2). Join corresponding root branch sets along these paths. Their interiors are disjoint; assigning each to its corresponding branch set preserves disjointness and connectivity. Again they give a K_7^- model. This excludes two heavy five-fragments. A heavy five-fragment and a positive fragment of boundary at most four contradict (1). Two positive fragments of boundary at most four contradict bilightness. $\square$

Lemma 5.3. *We have $e(G) = 4v(G) - 2$.*

Proof. If $\rho(G) \geq -1$, delete an edge uv to obtain J . Minimality forces a dense bifragment (S, T) in J , with both boundaries of order at most four. Unless this already violates bilightness in G , we may label the ends so that $u \in S$ and $v \notin S \cup N_J(S)$. If $v \in T$, choose the labels also so that $|N_G(S)| \geq |N_G(T)|$. Then $|N_G(S)| \leq 5$ and $\rho_G(S) = \rho_J(S) + 1 \geq 2$, so $|S| \geq 2$.

Put $S' = S \setminus \{u\}$. We have

$$N_G(S') \subseteq (N_G(S) \setminus \{v\}) \cup \{u\}, \quad \rho_G(S') = \rho_G(S) + 4 - e_G(\{u\}, N_G(S)).$$

The pair (S', T) is nonadjacent in G . If $|N_G(S)| \leq 4$, both members have boundary at most four and positive density, a contradiction. Otherwise Lemma 5.2(2) gives $\rho_G(S) \geq 7$, hence $\rho_G(S') \geq 6$. The density of T is positive, and at least two if its boundary grows to order five. Lemma 5.2(3) again gives a contradiction. $\square$

6. DEGREE BOUNDS AND FIVE-VERTEX CUTS

Lemma 6.1. *The graph G has minimum degree at least five and is five-connected.*

Proof. If $d_G(v) \leq 4$, the minor $J = G - v$ has density at least -2 and at least eight vertices. Any dense bifragment in J with boundaries of order at most four remains a bifragment in G . Its members remain positive; if a boundary grows to order five, the restored vertex adds at least one incident edge, so the density is at least two. Lemma 5.2(3) excludes such a pair. Thus J is 4-bilight, contrary to minimality.

Suppose that G has connectivity $k \leq 4$, and let S be a minimum cut, possibly empty. Each component of $G - S$ has neighbourhood S . By bilightness some component A has nonpositive density. The side R_A is internally k -connected, and its nonroots retain degree at least five. For

$k \leq 2$, its connected interior gives a rooted clique. For $k = 3$, apply Lemma 3.2; for $k = 4$, apply Lemma 3.3. In the last case $|A| \geq 2$, since the minimum degree is five. Another component together with S has at least six vertices, again by the degree bound. Hence A is reducible, contrary to Lemma 5.1. $\square$

Write $t(xy) = |N_G(x) \cap N_G(y)|$ for an edge xy . Contracting this edge changes the global density by

$$(6) \quad \rho(G/xy) = \rho(G) + 3 - t(xy).$$

A quasi-five-connected graph is 4-bilight: any member of a dense bifragment has boundary four and at least two vertices, and the other member supplies at least two vertices on the opposite side of that cut. Thus a quasi-five-connected contraction with $t(xy) \leq 3$ would contradict minimality.

We will repeatedly use the following observation. Contracting an edge of a five-connected graph gives a four-connected graph, and every four-cut contains the merged vertex. Consequently, in a dense bifragment of the contraction with boundaries of order at most four, the merged vertex belongs to both boundaries. Its open sides lift to positive five-fragments of the original graph, with both contracted ends in each boundary. Their densities do not decrease; they remain disjoint and nonadjacent. Each has at least two vertices, because a singleton of boundary four has density zero in the contraction.

Lemma 6.2. *The minimum degree of G is at least six.*

Proof. Suppose that $d_G(v) = 5$. If $N_G(v)$ is a clique, choose a component C outside $N_G[v]$. Such a component exists because $v(G) \geq 9$, and five-connectivity gives $N_G(C) = N_G(v)$. The five singleton neighbours, $\{v\}$ and C form a K_7^- model.

Otherwise choose nonadjacent $a, b \in N_G(v)$ and set $J = G - v + ab$. This is a minor: contract va , then delete the new edges at a other than ab . It has density -2 , so a dense bifragment (Y, Z) with boundaries of order at most four exists in J . For either member, say Y , put $k = e_G(\{v\}, Y)$ and let c be one if the added edge ab has an end in Y , and zero otherwise. Then

$$(7) \quad \rho_G(Y) = \rho_J(Y) + k - c.$$

Since $c = 1$ implies $k \geq 1$, both sets remain positive in G ; they are still disjoint and nonadjacent, and their boundaries have order at most five. By Lemma 5.2(3), we may choose Y with boundary five and $\rho_G(Y) = 1$. Its boundary grows on restoring v , so $k \geq 1$. Equation (7) forces $\rho_J(Y) = k - c = 1$. Exchange a, b if necessary: a is the unique neighbour of v in Y , and b lies in the boundary of Y in both graphs. Hence $|Y| \geq 2$. The opposite side of Y in G contains Z , which has at least two vertices.

If $|Y| \geq 3$, Lemma 3.4 makes G/va quasi-five-connected. As ab is absent and $d_G(v) = 5$, we have $t(va) \leq 3$, a contradiction. We are left with $Y = \{a, c\}$. The equation

$$1 = d_G(a) + d_G(c) - e(G[\{a, c\}]) - 8$$

forces $ac \in E(G)$ and $d_G(a) = d_G(c) = 5$. Their common neighbours form a triple R , and their complete neighbourhoods are

$$(8) \quad N_G(a) = \{c, v\} \cup R, \quad N_G(c) = \{a, b\} \cup R, \quad |R| = 3.$$

In particular vb is an edge.

Contract ac , which has exactly three common neighbours. By minimality and the preceding contraction observation, a dense bifragment lifts to two positive five-fragments. Choose a member T of density one, using Lemma 5.2(3). Lifting increases its density by $|T \cap R|$, so T avoids R . Both a, c are in its boundary; thus $v, b \in T$, and v is the unique neighbour of a in T . Its opposite side has at least two vertices. If $|T| \geq 3$, Lemma 3.4 applies to av ; the missing edge cv gives $t(av) \leq 3$, again a contradiction. Therefore $T = \{v, b\}$, and the same degree calculation gives

$$(9) \quad N_G(v) = \{a, b\} \cup D, \quad N_G(b) = \{c, v\} \cup D, \quad |D| = 3.$$

The four degree-five vertices induce the cycle a, c, b, v, a , and R, D avoid the cycle.

If $|R \cap D| \geq 2$, the cycle has at most four external neighbours. There is a vertex outside the cycle and its boundary, because $v(G) \geq 9$, contradicting five-connectivity. If $|R \cap D| = 1$, put $S = R \cup D$ and $U = \{a, c, b, v\}$. Here $|S| = 5$ and $\rho_G(U) = 0$. Its opposite side W has

$$\rho_G(W) = 18 - e(G[S]) \geq 8.$$

In particular W is nonempty. Its closed side rooted at S is internally five-connected, so the rooted clique theorem applies. The connected branch sets $\{a, v\}$ and $\{c, b\}$ are adjacent to each other and to every vertex of S . Together with the rooted clique they give a K_7 model.

Finally suppose that $R \cap D = \emptyset$. Every five-cut L containing a, v must also contain c, b . Indeed, if $c \notin L$, all of R lies in its component of $G - L$ or in L . Then a has no neighbour in any other component, and removing a from L leaves a cut of order four. The argument for b is the same, using the neighbours of v .

Now $t(av) = 0$. The contraction G/av satisfies the density bound, so it has a dense bifragment. Both lifted five-boundaries contain all four cycle vertices, by the preceding paragraph. Each open side meets both R and D , because its boundary contains a, v . One of these disjoint sides, say B , contains exactly one vertex r of R . It cannot be a singleton, since its image has positive density and boundary four. If $|B| = 2$, then $B = \{r, d\}$ for some $d \in D$, and its boundary consists of the four cycle vertices and one further vertex z . The vertex r is adjacent to neither v nor b , so its neighbours would all lie in $\{a, c, d, z\}$, contrary to the minimum degree five.

Thus $|B| \geq 3$, its opposite side has at least two vertices, and a has the unique neighbour r in B . Lemma 3.4 makes G/ar quasi-five-connected. Also $t(ar) \leq 3$, since a has degree five and its neighbour v is not adjacent to r . This final contraction contradicts minimality. Every contraction above leaves at least eight vertices, preserving the order condition. $\square$

Lemma 6.3. *Every proper low five-fragment A contains an endpoint of an edge e with $t(e) \leq 3$.*

Proof. Put $S = N_G(A)$ and $a = |A|$. The minimum degree six gives

$$\rho_G(A) = \sum_{v \in A} d_G(v) - e(G[A]) - 4a \geq 2a - \binom{a}{2},$$

so $a \geq 5$. Also

$$\sum_{v \in A} (d_G(v) + |N_G(v) \cap S|) = 8a + 2\rho_G(A) \leq 8a + 2.$$

Choose $v \in A$ whose summand is at most eight. Suppose that every edge at v has at least four common neighbours. Let H be $G[N_G(v)]$ with the edges within $X_0 = N_G(v) \cap S$ deleted. Vertices of $H - X_0$ have degree at least four, and vertices in X_0 have degree at least $5 - |X_0|$. Moreover $v(H) + |X_0| \leq 8$.

The side $G[A \cup S]$, rooted at S , is internally five-connected. Choose five disjoint paths from S to $N_G(v)$, stopped when they first reach $N_G(v)$. They avoid v , and the paths at vertices of X_0 are trivial. Their five ends form a set containing X_0 . Apply Lemma 3.1 to H rooted at these ends. Its central branch set avoids S . Use this branch set and $\{v\}$ as helpers. Four linkage paths become root branch sets. For the fifth, take the path ending in the central branch set and delete its last vertex. The trimmed path is nonempty, since its original end lies outside X_0 and hence outside S . Both helpers are adjacent to the other four root branch sets and to each other, and the trimmed branch set is adjacent to the central helper. Thus at least ten of the eleven helper adjacencies are present. Every branch set lies in $A \cup S$, and no artificial boundary edge has been used.

The opposite side is internally five-connected and has density at least $m + 7$, where $m = 10 - e(G[S])$, by Lemma 5.2(2). It supplies a rooted clique on S . Joining corresponding root branch sets gives a K_7^- model, a contradiction. $\square$

The following elementary argument will exclude the remaining five-cuts. A proper five-fragment is called an F -fragment, for a family F of edges, if its boundary contains both ends of an edge in F . An F -end is an F -fragment containing no F -fragment as a strict subset.

Lemma 6.4. *Let J be five-connected, with every proper five-fragment of order at least four. If A is an F -end, no edge $e \in F$ can meet A , lie in $A \cup N_J(A)$ and have both ends in a five-cut of J .*

Proof. Put $S = N_J(A)$ and $C = V(J) \setminus (A \cup S)$. Suppose a five-cut T contains both ends of such an edge e , and partition $J - T$ into two nonempty unions of components B, D . Write

$$\begin{aligned} p &= |A \cap T|, & s &= |S \cap T|, & r &= |C \cap T|, \\ u &= |S \cap B|, & w &= |S \cap D|. \end{aligned}$$

Then $p + s + r = u + s + w = 5$, $p \geq 1$ and $p + s \geq 2$.

If $A \cap B \neq \emptyset$, its neighbourhood lies in $(A \cap T) \cup (S \cap T) \cup (S \cap B)$. This set has order at least five, by connectivity. If it has order five, it equals the boundary and contains both ends of e , giving an F -fragment strictly inside A . Its opposite side contains C , and strict containment follows from $p \geq 1$. Therefore

$$p + s + u \geq 6, \quad p > w, \quad u > r.$$

The opposite corner $C \cap D$ has boundary contained in a set of order $r + s + w = 5 + r - u < 5$, so it is empty. Symmetrically, if $A \cap D \neq \emptyset$, then $p > u$, $w > r$ and $C \cap B$ is empty. In each application, a nonempty opposite corner ensures that the separation is proper.

If both A -corners are nonempty, then $C \subseteq T$, so $r = |C| \geq 4$, contrary to $p + s \geq 2$. If just $A \cap B$ is nonempty, then $D = S \cap D$, giving $w = |D| \geq 4$. Since $p > w$, we have $p = 5$ and $r = s = 0$. Now $C = C \cap B$ has all its neighbours in $S \cap B$, of order $u = 5 - w \leq 1$, again impossible. The other one-corner case is symmetric.

If neither A -corner is nonempty, then $A \subseteq T$ and $p = |A| \geq 4$. Since $|C| \geq 4$ and $r \leq 1$, a C -corner is nonempty, say $C \cap B$. Its boundary has order at most $r + s + u$, so connectivity gives $u \geq p \geq 4$. The other C -corner must be empty, or similarly $w \geq 4$. But then $D = S \cap D$, and again $w = |D| \geq 4$. This contradicts $u + w \leq 5$. $\square$

Lemma 6.5. *The graph G is six-connected.*

Proof. It is already five-connected. We first show that every proper five-fragment has at least four vertices. A singleton would have degree five. For a fragment of order $a = 2$ or 3 , the degree bound and the possible incident edges give

$$2a - \binom{a}{2} \leq \rho_G(A) \leq \binom{a}{2} + a.$$

Its density is therefore three if $a = 2$, and between three and six if $a = 3$. Both contradict Lemma 5.2(2), which allows only densities at most one or at least seven. Also, every low five-fragment is proper: an empty opposite side would give density $18 - e(G[N_G(A)]) \geq 8$.

Let F be the edges with at most three common neighbours. Every edge in F has both ends in the boundary of a five-fragment of density one. Indeed, its contraction preserves the density bound by (6), so is not 4-bilight. Its dense bifragment lifts to two positive five-fragments, as described above; Lemma 5.2(3) forces one of their densities to equal one.

Suppose a five-cut exists. It has a low side, so Lemma 6.3 supplies an edge of F . The preceding paragraph supplies a low F -fragment. Choose an inclusion-minimal one, A . In fact A is an F -end without any density restriction. For a smaller F -fragment $B \subsetneq A$, a heavy B would be disjoint and nonadjacent to the heavy opposite side of A , violating Lemma 5.2(3). Thus B would be low, contradicting the choice of A .

Lemma 6.3 now supplies an edge $e \in F$ meeting A . Its other end lies in $A \cup N_G(A)$, and both ends lie in a five-cut by the earlier contraction argument. This contradicts Lemma 6.4 and excludes every five-cut. $\square$

7. THE FINAL DEGREE CASES

The degree-six argument adapts the final part of the proof of Norin and Totschnig in [6, Section 3]; six-connectivity excludes the separator outcome directly. It uses the two-paths theorem of Robertson, Seymour and Thomas [7, Theorem (2.4)], in the form also quoted in [6,

Theorem 13]. For four distinct vertices in a specified cyclic order, it gives disjoint paths joining opposite pairs, a separation of order at most three with all terminals on one side and at least two vertices in the other open side, or a drawing in a disc with the terminals on its boundary in that order.

Lemma 7.1. *Let H be six-connected, with at least nine vertices and no K_7^- minor. If every edge has at least four common neighbours and H has a vertex of degree six, then $e(H) \leq 4v(H) - 9$.*

Proof. Let $d_H(v) = 6$. The complement of $H[N_H(v)]$ is a matching, since each neighbour of v has at least four neighbours there. Label the six neighbours $u_1, w_1, u_2, w_2, u_3, w_3$ so that the only possible nonedges are $u_i w_i$.

There cannot be disjoint paths joining two different pairs u_i, w_i with interiors outside $N_H[v]$: contracting the interior of each path into one of its ends supplies two of the three possibly missing adjacencies. Together with $\{v\}$, the six resulting branch sets give K_7^- . In particular all three pairs are nonedges. Otherwise an edge joining one pair and a path for another pair through an exterior component would suffice. Such a component exists and is adjacent to all six neighbours of v , by six-connectivity.

Some pair, say u_1, w_1 , has no common neighbour outside $N_H[v]$. Otherwise choose an exterior common neighbour x_i for each pair. Two distinct choices give the forbidden disjoint length-two paths, so all choices coincide at a vertex x . There is a component C outside $N_H[v] \cup \{x\}$, because $v(H) \geq 9$. Its boundary is contained in $N_H(v) \cup \{x\}$ and has at least six vertices. Thus at least five neighbours of v meet C , giving a path for one pair through C and a disjoint path for another through x .

Apply the two-paths theorem to $H' = H - \{v, u_1, w_1\}$, with ordered terminals u_2, u_3, w_2, w_3 . The two paths are excluded above. A separation (A, B) in the second outcome lifts to

$$(A \cup \{v, u_1, w_1\}, B \cup \{u_1, w_1\})$$

in H , of order at most five. It is proper: v lies in the first open side, and the second contains $B \setminus A$. No neighbour of v lies in that second open side. This contradicts six-connectivity.

Hence H' has the disc drawing from the third outcome. Its four distinct boundary terminals give $e(H') \leq 3v(H') - 7$. Every vertex outside $N_H[v]$ meets the deleted triple at most once, by the choice of u_1, w_1 . Each of the other four neighbours of v meets it three times, and the triple induces two edges. With $n' = v(H') = v(H) - 3$,

$$e(H) \leq (3n' - 7) + (n' - 4) + 12 + 2 = 4v(H) - 9.$$

□

Proof of Theorem 2.1. The minimum counterexample G is six-connected by Lemma 6.5. Every edge has at least four common neighbours: otherwise contracting it gives a five-connected, hence 4-bilight, smaller graph of density at least -2 . The average degree is $8 - 4/v(G) < 8$, whereas Lemma 6.2 gives minimum degree at least six. Thus some vertex has degree six or seven.

Degree six contradicts Lemma 7.1, since $e(G) = 4v(G) - 2$. For a degree-seven vertex v , put $J = G[N_G(v)]$. The common-neighbour bound gives $\delta(J) \geq 4$. Choose a component C outside $N_G[v]$, which exists since $v(G) \geq 9$. Its boundary lies in $N_G(v)$ and has at least six vertices. Contract C to a vertex and delete the other exterior components. The resulting nine-vertex minor consists of J , the vertex v adjacent to all of J , and a nonadjacent second vertex adjacent to at least six vertices of J . Lemma A.1 gives a K_7^- model in this minor. Replacing the contracted vertex by the connected set C lifts the model to G .

Both degree cases are impossible, so no counterexample exists. □

Proofs of Corollary 1.2 and Theorem 1.1. Every five-connected graph is 4-bilight, giving Corollary 1.2. For the colouring theorem, suppose a K_7^- -minor-free graph is not six-colourable, and choose such a graph G minor-minimal. Then its chromatic number is at least seven and every proper minor is six-colourable. The critical-graph reduction [2, Theorem 1.6] gives seven-connectivity and $e(G) \geq 4v(G) - 2$. The graph is therefore 4-bilight, and Theorem 2.1 supplies the forbidden minor. □

APPENDIX A. THE DEGREE-SEVEN QUOTIENT LEMMA

Lemma A.1. *Let R be a graph on seven vertices with minimum degree at least four. Add a vertex v adjacent to every vertex of R , and another vertex c adjacent to at least six vertices of R . No edge vc is required. The resulting graph contains a K_7^- minor.*

Proof. Delete edges if necessary so that vc is absent and c misses exactly one vertex x of R . The complement F of R has maximum degree at most two. Add edges to F until it is edge-maximal subject to this bound, obtaining F' . Replacing R by $\overline{F'}$ only deletes edges, so it suffices to find the minor in the resulting graph.

Every component of F' is a path or a cycle. A path of order at least three could be closed into a cycle, and two distinct path components could be joined at vertices of degree less than two. Maximality therefore leaves only cycles and at most one additional K_1 or K_2 . Since F' has seven vertices, its possible forms are

$$C_7, \quad C_4 \dot{\cup} C_3, \quad C_6 \dot{\cup} K_1, \quad C_3 \dot{\cup} C_3 \dot{\cup} K_1, \quad C_5 \dot{\cup} K_2.$$

Label each cycle consecutively as indicated below. Up to automorphisms of F' , there are nine choices of the marked vertex x . The table gives a model for each; two rows each cover two choices. Concatenated labels denote a branch set, for example $03 = \{0, 3\}$ and $2v = \{2, v\}$.

F'	x	seven branch sets	only missing pair
$C_7(0, \dots, 6)$	0	$03, 15, 2, 4, 6, v, c$	$v : c$
$C_4(0, \dots, 3) \dot{\cup} C_3(4, 5, 6)$	0, 4	$04, 1, 25, 3, 6, v, c$	$v : c$
$C_6(0, \dots, 5) \dot{\cup} K_1(6)$	0	$024, 1, 3, 5, 6, v, c$	$v : c$
$C_6(0, \dots, 5) \dot{\cup} K_1(6)$	6	$02, 1, 3, 4c, 5, 6, v$	$02 : 1$
$C_3(0, 1, 2) \dot{\cup} C_3(3, 4, 5) \dot{\cup} K_1(6)$	0	$03, 14, 2, 5, 6, v, c$	$v : c$
$C_3(0, 1, 2) \dot{\cup} C_3(3, 4, 5) \dot{\cup} K_1(6)$	6	$03, 1c, 2, 4, 5, 6, v$	$4 : 5$
$C_5(0, \dots, 4) \dot{\cup} K_2(5, 6)$	0, 5	$05, 1, 2v, 3, 4, 6, c$	$3 : 4$

In every row the branch sets are disjoint and connected, and all pairs except the indicated pair are adjacent. For example, in the first row 03 and 15 are edges of $\overline{F'}$; together with the singleton branch sets $2, 4, 6$, they are pairwise adjacent. Both v and c are adjacent to all five sets. The other rows are checked in the same way. Contracting the branch sets gives a K_7^- minor. $\square$

USE OF AI

OpenAI models helped develop the proof and assisted with computation, drafting and internal proof checks. The author guided the research plan.

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